

Lecture 12 - Stability Margins

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Stability Margins: How much we can change control gains before instability.

- Describe a physical system and highlight model uncertainty to motivate stability margins.
- Stability margins are an intuitive way to find gain and phase margins using Bode plots.

12.1 Position Control of a DC Motor

$$\underbrace{T}_{\text{Torque}} - \underbrace{b}_{\text{friction}} \dot{\theta} = \underbrace{I}_{\text{Inertia}} \ddot{\theta} \quad \left\{ \begin{array}{l} \theta = \text{position} \\ \dot{\theta} = \text{velocity} \\ \ddot{\theta} = \text{acceleration} \end{array} \right.$$
$$\rightarrow I\ddot{\theta} + b\dot{\theta} = T$$
$$\rightarrow G(s) = \frac{\Theta(s)}{T(s)} = \frac{Y(s)}{U(s)} = \frac{1}{Is^2 + bs} = \frac{1}{s(Is + b)} = \frac{1/I}{s(\frac{I}{I}s + \frac{b}{I})} = \frac{A}{s(s + c)}$$

Now we have a model of our plant G in a familiar form.

Q: Our new variable A and c are based on Inertia I and friction term b

→ Are we 100% confident in our ability to model values for these?

Ans: No!

→ Let's see what else might contribute to uncertainty about the system.

Block diagram

Let's consider our standard block diagram using a controller H to control G, and label the signals.

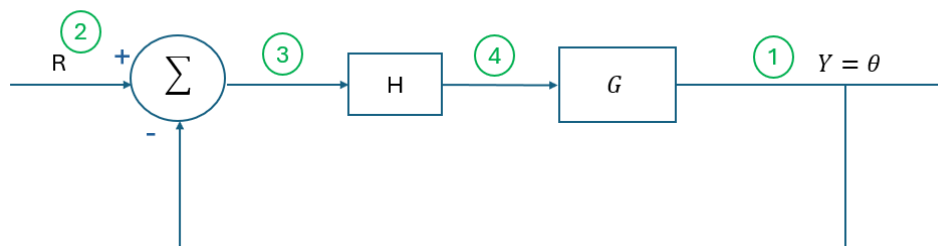


Figure 12.1: Standard block diagram with reference R, controller H, plant G, and output Y.

Q: What are the signals representing?

1. $Y = \theta \leftarrow$ An angle position (measured). E.g. read by encoder mounted to our motor.
2. $R \leftarrow$ Also an angle (desired)
3. $e \leftarrow$ Angle error $R - Y$
4. $? = T = U(s) \leftarrow$ A torque, or control input. $G = \frac{\theta}{T}$ i.e. maps torque to angle.

Q: Our controller H is just an algorithm on a computer/micro-controller, so how does it exert a torque on the motor?

→ Lets modify our block diagram to understand what H actually outputs in this example and why that generates a torque.

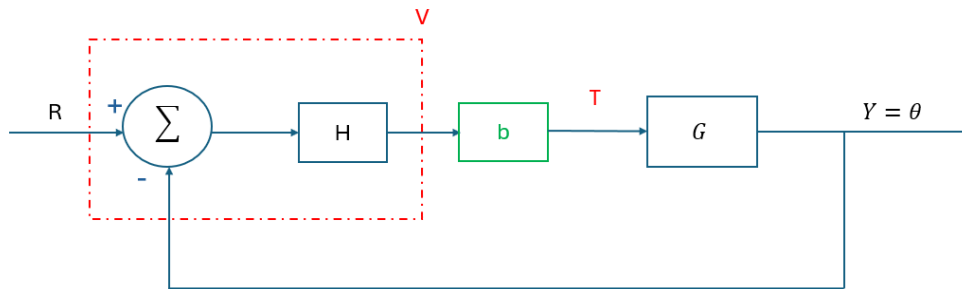


Figure 12.2: Block diagram with gain b added.

Let's add gain b: we usually just set this to 1 so we can ignore it and focus on key concepts, but it will help illustrate other uncertainties.

Let's assume we're controlling our DC motor using an Arduino Uno.

→ Arduino is a popular micro-controller that can be

1. Connected to our encoder to measure angle Y (and compare to R)
2. Connected to our motor, and can set a voltage to drive it with.

Disclaimer: Physical implementation is a little more complex, e.g. PWM, drive using H-bridge.

Gain b needs to map voltage V to torque T:

$$\begin{aligned}
 \rightarrow \text{Torque } T &= k_i \cdot i, & \text{where } i \text{ is the current and } k_i \text{ is the torque constant from motor datasheet} \\
 \rightarrow \text{Current } i &= \frac{V}{R} \\
 \rightarrow T &= \frac{k_i}{R} \cdot V \quad \rightarrow \quad b = \frac{k_i}{R} \quad \} \text{ Our model for } b
 \end{aligned}$$

We've previously discussed how controller gains K are capped due to "actuator saturation". In this example, we're limited by the motor stall torque, or even by the max current we have available.

Q: Are we 100% confident we can measure/rely on these values?

Ans: No!

- k_i will have tolerance and be a linear approximation of T and i relationship.
- R will change, even with room temperature and particularly when current is applied.

∴ Because of uncertainties like these, classical control theorists introduced stability "margins" when designing controllers to assess how much gain and phase H can add to a system before it goes unstable.

12.2 Stability Margins

12.2.1 Gain Margin and Phase Margin

Consider a closed loop transfer function

$$G_{RY} = \frac{GH}{1+GH}, \quad \text{where } H(s) = K \quad (\text{Proportional feedback})$$

$$= \frac{GK}{1+GK}$$

A transition to instability occurs if any LHP poles move to the RHP, or if gain of $G_{RY} \rightarrow \infty$
i.e. if $1+GK = 0 \implies G_{RY} \rightarrow \infty$

→ We must avoid $GK = -1$

Q: What does this mean in terms of gain and phase of GK?

$$\text{Instability occurs when: } \left. \begin{array}{l} |G(j\omega)k| = 1 \\ \text{AND} \\ \angle G(j\omega)k = -180^\circ \end{array} \right\} \text{ if Bode plot has: } \begin{array}{l} 0 \text{ dB gain} \\ \text{AND} \\ -180^\circ \text{ phase} \end{array}$$

→ Gain and phase margins tell us how far we are from this.

Example

Consider G(s) for plant,

$$G(s) = \frac{1}{s(s+1)(s+10)}$$

We want to know:

- How much gain and phase margin do we have?
- How large k_p can we apply before the system goes unstable?

First, draw Bode plot, consider G in 2 parts:

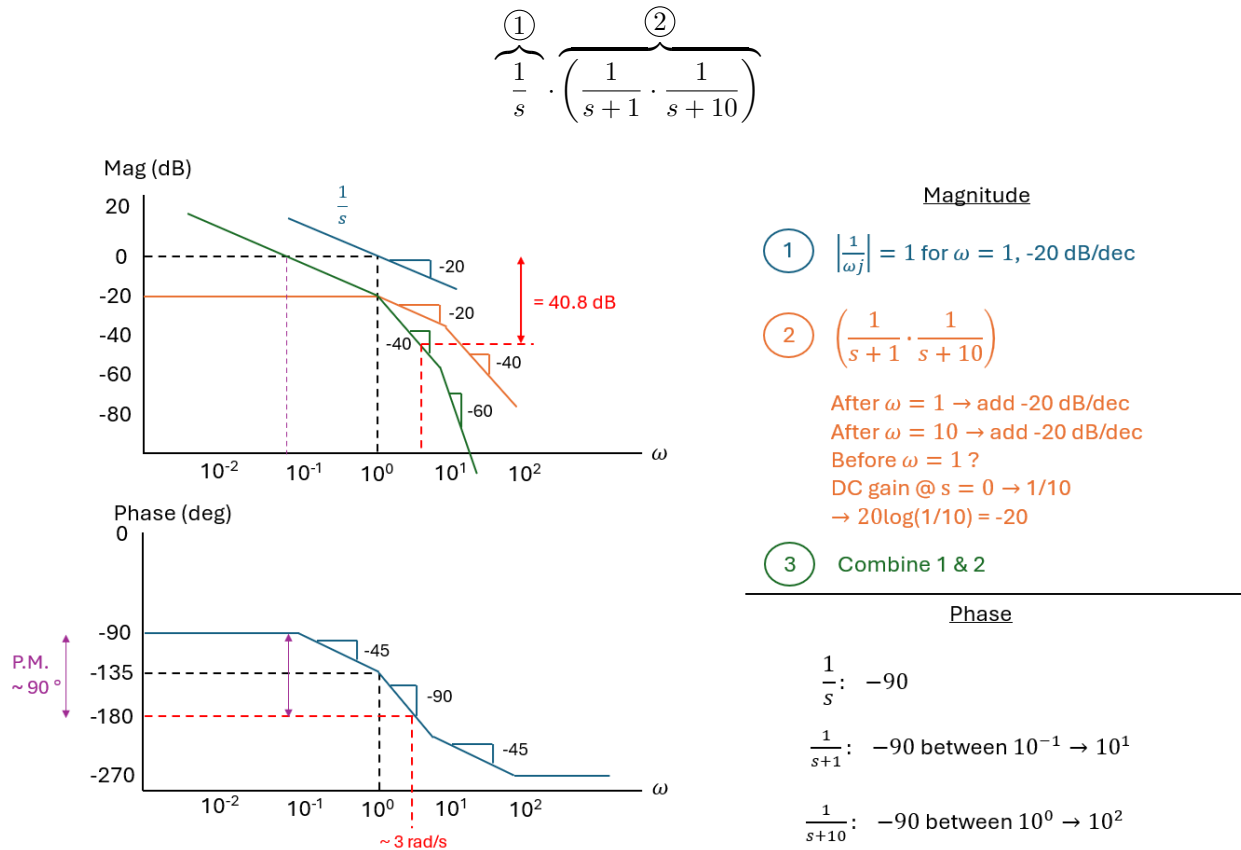


Figure 12.3: Using the Bode plot to determine the gain and phase margin.

Now that we have our Bode plot, we can find our gain and phase margins.

12.2.2 Gain Margin GM

1. Identify the "Phase crossover frequency"

$\rightarrow$ Where the phase crosses $-180^\circ \Rightarrow \approx 3 \text{ rad/s}$

2. The gain margin is the margin between 0 dB and the magnitude of G at the phase crossover frequency.

$\rightarrow GM \approx 40.8 \text{ dB}$

$\rightarrow$ If we add $\approx 40 \text{ dB}$ (or more) of controller gain K, our system will be unstable. $\leftarrow$ We are essentially amplifying a signal with 180° phase shift: positive feedback.

What's the max K we can use?

$$20\log_{10}(K) = 40.8$$

$$\rightarrow \log_{10}(K) = 2.04$$

$$\Rightarrow K = 10^{2.04} = 110$$

Recall multiplying by gain will move the whole magnitude curve up, but not affect phase.

12.2.3 Phase Margin

1. Identify the "gain crossover frequency"

→ Where magnitude crosses 0 dB $\implies \omega \approx 10^{-1}$

2. Phase margin is margin between -180° and the phase of G at the gain crossover frequency

→ PM $\approx 90^\circ$

→ If we add 90° or more phase, our system will go unstable.

Q: Do we know any other ways to place conditions on gain K for stability?

Ans: Routh Hurwitz:

$$H(s) = K, \quad G(s) = \frac{1}{s(s+1)(s+10)}$$

$$G_{RY} = \frac{\frac{K}{s(s+1)(s+10)}}{1 + \frac{K}{s(s+1)(s+10)}} = \frac{K}{s(s^2 + 11s + 10) + K} = \frac{K}{s^3 + 11s^2 + 10s + K}$$

$$a_3 > 0 \quad \rightarrow \quad K > 0$$

$$a_1 \cdot a_2 > K \quad \rightarrow \quad 110 > K \quad \leftarrow \quad \text{Same as we found for gain margin!}$$

So why do we care about gain and phase margins?

- Many tools in classical control theory help us evaluate stability in different ways, including Routh Hurwitz, gain and phase margin using Bode, and Nyquist plots (next week).
- Routh Hurwitz gives us bounds on K for stability, but using stability margins on Bode plots gives us a graphical way to easily meet design goals.
- Because of uncertainties in our model, control engineers can give design requirements specifying a minimum gain and phase margin.

Example: Choose K to give a gain margin of 20 dB

$$\underbrace{40.8}_{\text{old GM}} - \underbrace{20}_{\text{new GM}} = 20.8 \text{ dB}$$

$$\rightarrow 20\log(K) = 20.8$$

$$\implies K = 10^{1.04} = 11$$